

Quantized Model Soup Shake-Up: Weight Perturbation for Enhanced Ensemble Diversity

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Abstract

Model soup, averaging the weights of multiple fine-tuned models, delivers ensemble-level accuracy at single-model inference cost, but its success requires both linear mode connectivity (LMC) and sufficient diversity among candidates. We study these two requirements under quantization-aware training (QAT). First, we show analytically and empirically that fine-tuning from a sufficiently converged QAT checkpoint, which we call the QAT anchor state, preserves linear mode connectivity by keeping models within the same loss basin. Second, we identify Ensemble Degeneracy, where the many-to-one mapping of the quantizer collapses independently fine-tuned models into nearly identical quantized representations, eliminating diversity despite intact connectivity. To resolve this, we propose **Quantized Model Soup Shake-Up (QMSS)**, which selectively perturbs low-magnitude weights near quantization bin boundaries to flip their integer assignments, then briefly re-trains each variant via QAT. Because only a small fraction of inherently unstable weights are modified, QMSS induces substantial quantized-domain diversity while preserving LMC. Experiments on CIFAR-100, Tiny-ImageNet, ImageNet, and FSD-Kaggle2018 with both EWGS and LSQ quantizers on CNNs and Vision Transformers show consistent improvements over standard model soup, with gains up to +1.54% Top-1 on CIFAR-100 and +2.09% on CIFAR-100-C, alongside over 6.37× inference speedup on a Jetson Orin Nano at 8/8-bit.

CCS Concepts

• **Computing methodologies** → **Machine learning; Neural networks; Ensemble methods.**

Keywords

quantization, model soup, ensemble

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1 Introduction

Deep neural networks have recently achieved impressive performance improvements across various AI applications, but at the cost of large model sizes and increased computational demands. Model quantization alleviates this by reducing the bit-precision of weights and activations, significantly lowering memory and compute requirements [21, 28]. In particular, quantization-aware training (QAT) enables low-bit models to retain full-precision accuracy by simulating quantization during training [9, 26].

Concurrently, model soup ensembling has emerged as a cost-effective alternative to output-level ensembles [40]. By averaging the weights of multiple fine-tuned models, it produces a single refined model without additional inference cost. Crucially, this approach relies on the assumption that the models lie in a connected region of the loss landscape, a property known as *linear mode connectivity* (LMC) [12, 13, 44], which ensures that linear interpolation between models does not degrade performance. To satisfy this condition, model soup is typically constructed by fine-tuning multiple models from the same pretrained initialization. Beyond LMC, recent theoretical advances in ensemble learning [33, 39] have established that ensemble performance depends critically on *diversity* among base models. The expected ensemble risk decomposes into a sum of individual risks and a diversity term, implying that both the individual performance of each model and their mutual diversity are essential for achieving superior ensemble performance. If all base models coincide, the ensemble provides no gain. Therefore, effective model soup in the quantized regime must simultaneously satisfy two requirements: *linear mode connectivity* and *sufficient diversity*.

However, quantization poses distinct threats to each of these requirements. A recent study on post-training quantization [34] has shown that quantization-induced perturbations can disrupt LMC by introducing interpolation barriers. In QAT, this risk is further amplified because both quantization parameters and weights evolve jointly during training, potentially driving models into disconnected loss basins. At the same time, even when connectivity

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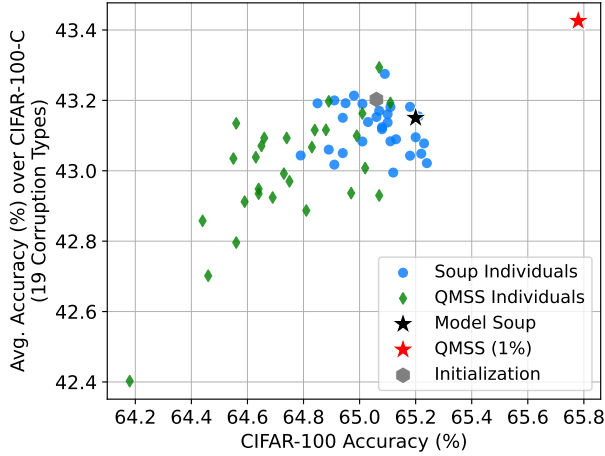


Figure 1: QMSS improves accuracy over standard model soup by introducing quantization-level diversity. Each dot represents a model fine-tuned from a shared QAT anchor state using random hyperparameter configurations. The red star denotes the QMSS soup obtained by averaging perturbed models (green), while the black star represents the standard model soup (blue). Despite slightly lower individual accuracy, the QMSS achieves better ensemble performance on both CIFAR-100 and CIFAR-100-C.

is preserved, the discrete nature of quantization can collapse independently fine-tuned models to nearly identical quantized representations, eliminating the diversity essential for ensemble gains.

We address both challenges in this work. For the first, we show that fine-tuning from a sufficiently converged QAT checkpoint, which we term the **QAT anchor state**, keeps all resulting models within a linearly connected region. Intuitively, once quantization parameters (e.g., clipping bounds) have stabilized, subsequent fine-tuning induces only small perturbations that remain within the same loss basin. We provide both an analytical framework and empirical verification of this finding.

For the second challenge, we identify a fundamental obstacle that we call **Ensemble Degeneracy** in QAT. The core of the problem is the many-to-one nature of the quantizer $Q(\cdot)$: quantization maps a continuous interval of real-valued weights to the same integer value. As a result, a small weight update during fine-tuning is often insufficient to shift a weight across a quantization bin boundary. Consequently, two fine-tuned QAT models θ_1 and θ_2 may satisfy $\|\theta_1 - \theta_2\|_2 > 0$ in full precision yet still obey $Q(\theta_1) = Q(\theta_2)$ for the vast majority of weights, collapsing to functionally identical quantized networks. Fig. 2 provides empirical evidence for this degeneracy: Model Soup exhibits low QWS and its gains saturate quickly as more models are averaged. To directly quantify diversity in the discrete weight space, we introduce *Quantized Weight Shift* (QWS), which measures the fraction of weights whose integer bin assignments differ between a candidate model and the weight-averaged soup. As shown in Fig. 2a, standard Model Soup exhibits remarkably low QWS, indicating that most weights remain in the

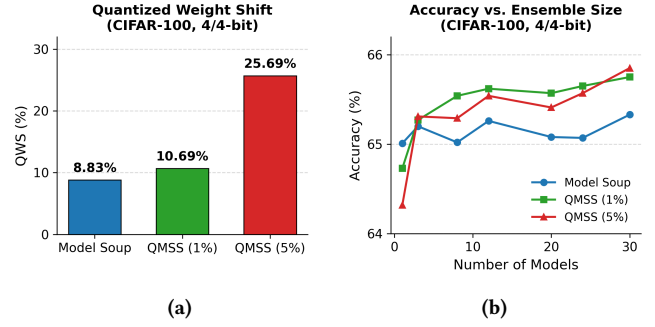


Figure 2: Results on CIFAR-100 with 4/4-bit quantization. (a) Quantized Weight Shift (QWS): higher values indicate greater diversity in the quantized weight space. QMSS substantially increases QWS compared to standard Model Soup. (b) Top-1 accuracy as a function of the number of averaged models. QMSS improves steadily with more candidates while standard Model Soup saturates earlier, resulting in consistently higher accuracy.

same quantization bin despite fine-tuning with different hyperparameters. Fig. 2b further shows that the soup accuracy saturates quickly as more models are averaged, yielding negligible ensemble gain. This is a direct consequence of the lack of quantized-domain diversity.

To overcome this degeneracy, we propose **Quantized Model Soup Shake-Up (QMSS)**. Our key insight is that *LMC and ensemble diversity can be simultaneously achieved* in the quantized regime by carefully perturbing only the inherently unstable weights near quantization boundaries. Specifically, QMSS selectively perturbs a small fraction of low-magnitude weights located within a narrow boundary zone around quantization thresholds, flipping their integer bin assignments. We target these weights for two reasons. First, boundary oscillation [27, 29] is the primary source of training-time bin-flipping instability, so these weights are already prone to change. Second, their small magnitudes indicate minimal contribution to the learned representations [11, 16], so perturbing them induces substantial quantized-domain diversity without disrupting the loss basin structure that underlies LMC. After perturbation, a brief QAT fine-tuning phase restores accuracy while preserving the newly introduced diversity. As shown in Fig. 1, although individual QMSS models suffer a slight accuracy drop due to the injected noise, the increased diversity leads to significantly stronger ensemble performance.

Previous attempts to ensemble low-precision models typically rely on output-level voting, snapshot ensembles, or Bayesian marginalization of quantization noise [30, 35, 36, 42, 46]. Although such approaches may improve robustness, they retain every individual model at inference time and thus cannot match the parameter efficiency of a merged weight soup. To our knowledge, prior work has not explicitly addressed these issues in the context of weight-space ensembling for QAT. In this work, we (i) demonstrate that LMC is preserved in QAT when fine-tuning from a QAT anchor state, (ii) diagnose the Ensemble Degeneracy problem that collapses quantized model diversity, and (iii) propose a principled remedy that enables

direct weight-space merging of QAT models. We summarize our key contributions as follows:

- We provide an analytical framework and empirical evidence showing that QAT models fine-tuned from a shared **QAT anchor state** maintain linear mode connectivity, through error-barrier analysis and interpolation experiments.
- We identify and formalize **Ensemble Degeneracy** in QAT, a fundamental obstacle where the many-to-one quantizer mapping collapses independently fine-tuned models into nearly identical quantized representations, rendering model soup ineffective despite preserved connectivity.
- We propose **QMSS**, a novel algorithm that resolves this tension by selectively perturbing low-magnitude weights near quantization thresholds, inducing substantial quantized-domain diversity while remaining within the same loss basin.
- Extensive experiments across diverse benchmarks and evaluation settings demonstrate consistent improvements and strong generalization across quantization schemes (EWGS, LSQ) and architecture families (CNNs and Vision Transformers), achieving gains up to +1.54% Top-1 on CIFAR-100 and +2.09% on CIFAR-100-C, alongside over 6.37× inference speedup on a Jetson Orin Nano at 8/8-bit.

2 Related Work

Model Quantization. Quantization methods fall into post-training quantization (PTQ) and quantization-aware training (QAT). PTQ calibrates a pre-trained model without retraining, while QAT jointly optimizes weights and quantization parameters, generally achieving higher accuracy at greater training cost. The core challenge of QAT is the non-differentiability of rounding, commonly handled by the straight-through estimator (STE). Subsequent works improve QAT through learnable step sizes (LSQ [9], LSQ+ [3]), adaptive gradient rescaling (EWGS [26]), and optimized clipping (QIL [23], PACT [6]).

Model Merging and Linear Mode Connectivity. Traditional ensembles aggregate predictions from multiple models [5, 39], but linearly increase inference cost. Parameter-space averaging methods such as Snapshot Ensembles [19], SWA [20], and Model Soup [40] produce a single merged model instead. Weight averaging succeeds only when models are linearly mode connected [1, 8], i.e., the loss stays low along their interpolation path. [31] showed that fine-tuned models remain within the pretrained loss basin, and [22] proposed repairing interpolation barriers via activation renormalization. While LMC is well studied in full precision, its behavior under quantization remains largely unexplored. [34] recently showed that post-training quantization can break connectivity via grid mismatch. We extend this line of analysis to QAT, showing that a shared QAT anchor state preserves LMC while the discrete quantizer collapses diversity.

Diversity in the Quantized Regime. Diversity is essential for effective ensembling [41], yet applying Model Soup to quantized models is nontrivial. The discrete nature of quantization maps slight weight variations to the same integer bin, causing independently trained

low-bit models to converge to highly similar representations. Simply varying random seeds or training schedules is therefore insufficient to induce meaningful diversity, limiting the benefits of weight-space merging in the quantized regime.

3 Analysis

Model soup via weight averaging is reliable only when the loss remains low along the straight line segment between candidate solutions. We analyze this via the *error barrier* and show (i) why linear mode connectivity (LMC) is expected in full precision when solutions stay in the same local basin, and (ii) how quantization can amplify this barrier. Finally, we argue that once QAT reaches a shared **QAT anchor state**, fine-tuning from the anchor stays within the same basin, thereby preserving LMC in practice.

3.1 Full-Precision Error Barrier and Linear Mode Connectivity

Let $L(\theta)$ denote the (validation) loss of a model with parameters $\theta \in \mathbb{R}^d$. Given two converged solutions θ_1, θ_2 , define their linear interpolation:

$$\theta_\lambda = (1 - \lambda)\theta_1 + \lambda\theta_2, \quad \lambda \in [0, 1]. \quad (1)$$

Error barrier. We quantify linear mode connectivity using the *error barrier*:

$$\mathcal{B}(\theta_1, \theta_2) = \max_{\lambda \in [0, 1]} \left[L(\theta_\lambda) - \frac{1}{2}(L(\theta_1) + L(\theta_2)) \right]. \quad (2)$$

A small $\mathcal{B}(\theta_1, \theta_2)$ indicates that the entire straight segment stays in a low-loss region, i.e., the two solutions are *linearly mode connected*.

A local basin view. Suppose both solutions are obtained by fine-tuning from a common reference point θ_0 (e.g., a pretrained checkpoint or a QAT anchor state):

$$\theta_1 = \theta_0 + \Delta_1, \quad \theta_2 = \theta_0 + \Delta_2. \quad (3)$$

Let $\Delta_\lambda = (1 - \lambda)\Delta_1 + \lambda\Delta_2$, so that $\theta_\lambda = \theta_0 + \Delta_\lambda$.

Assume L is twice differentiable in a neighborhood of θ_0 and let $H_0 = \nabla^2 L(\theta_0)$. A second-order Taylor expansion around θ_0 yields

$$L(\theta_\lambda) = L(\theta_0) + \nabla L(\theta_0)^\top \Delta_\lambda + \frac{1}{2} \Delta_\lambda^\top H_0 \Delta_\lambda + R_3(\Delta_\lambda), \quad (4)$$

where $R_3(\Delta_\lambda)$ is the third-order remainder.

If θ_0 is *sufficiently converged* (hence $\|\nabla L(\theta_0)\|$ is small) and fine-tuning stays local (hence $\|\Delta_\lambda\|$ is small), the quadratic term dominates. Using $\Delta_\lambda^\top H_0 \Delta_\lambda \leq \lambda_{\max}(H_0) \|\Delta_\lambda\|_2^2$, we obtain the standard bound:

$$\mathcal{B}(\theta_1, \theta_2) \lesssim \frac{\lambda_{\max}(H_0)}{2} \max_{\lambda \in [0, 1]} \|\Delta_\lambda\|_2^2 + \max_{\lambda \in [0, 1]} |R_3(\Delta_\lambda)|. \quad (5)$$

In particular, if the interpolation segment remains inside a single smooth *loss basin* around θ_0 , then $\|\Delta_\lambda\|$ is controlled and the barrier stays small, explaining why LMC often holds for fine-tuned models that do not drift far from a shared origin.

3.2 Quantization-Induced Barrier Amplification and QAT Anchor State

We now analyze the error barrier under quantization. Let $Q(\cdot)$ be a (uniform) quantizer. A common approximation models quantization

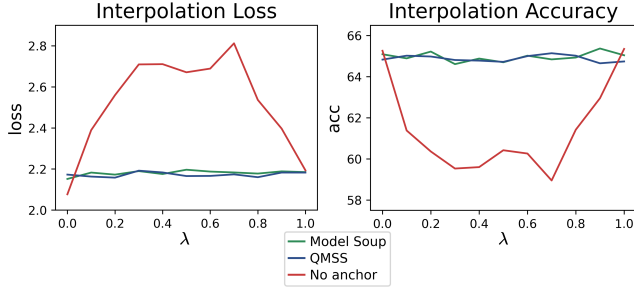


Figure 3: Linear mode connectivity from a QAT anchor state (ResNet-32, CIFAR-100, 4/4-bit). Loss (left) and accuracy (right) along the linear interpolation path $\theta_\lambda = (1 - \lambda)\theta_1 + \lambda\theta_2$. Models fine-tuned from a shared QAT anchor (Model Soup, QMSS) exhibit flat curves with near-zero barriers, confirming linear mode connectivity. QMSS preserves connectivity despite weight perturbations, as only a small fraction of low-magnitude weights are modified. In contrast, *No anchor* models—*independent* QAT runs from the same FP pretrained checkpoint without a shared anchor—show a pronounced loss barrier and accuracy drop near $\lambda = 0.5$, indicating convergence to different basins.

as additive noise:

$$Q(\theta_i) \approx \theta_i + \xi_i, \quad \xi_i \sim \mathcal{U}\left[-\frac{s_i}{2}, \frac{s_i}{2}\right]^d, \quad (6)$$

where s_i is the effective quantization step size of run i (possibly layer-wise).

Quantized error barrier. Define $\bar{\theta}_i = Q(\theta_i) \approx \theta_i + \xi_i$ and the quantized interpolation $\bar{\theta}_\lambda \approx \theta_\lambda + \xi_\lambda$. The quantized error barrier is

$$\mathcal{B}_Q(\theta_1, \theta_2) = \max_{\lambda \in [0,1]} \left[L(\theta_\lambda + \xi_\lambda) - \frac{1}{2}(L(\theta_1 + \xi_1) + L(\theta_2 + \xi_2)) \right]. \quad (7)$$

Expanding each loss term to second order around the corresponding full-precision point (e.g., around θ_λ) gives the decomposition

$$L(\theta_\lambda + \xi_\lambda) = L(\theta_\lambda) + \nabla L(\theta_\lambda)^\top \xi_\lambda + \frac{1}{2} \xi_\lambda^\top H_\lambda \xi_\lambda + r_\lambda, \quad (8)$$

$$L(\theta_i + \xi_i) = L(\theta_i) + \nabla L(\theta_i)^\top \xi_i + \frac{1}{2} \xi_i^\top H_i \xi_i + r_i, \quad i \in \{1, 2\}, \quad (9)$$

where $H_\lambda = \nabla^2 L(\theta_\lambda)$, $H_i = \nabla^2 L(\theta_i)$, and r_λ, r_i are higher-order remainders.

Plugging (8)–(9) into (7) and using the subadditivity of max yields the upper bound

$$\begin{aligned} \mathcal{B}_Q(\theta_1, \theta_2) \lesssim \mathcal{B}(\theta_1, \theta_2) + \max_{\lambda \in [0,1]} \left[\nabla L(\theta_\lambda)^\top \xi_\lambda + \frac{1}{2} \xi_\lambda^\top H_\lambda \xi_\lambda - \right. \\ \left. \frac{1}{4} (\xi_1^\top H_1 \xi_1 + \xi_2^\top H_2 \xi_2) \right] + O(\|\xi\|^3). \end{aligned} \quad (10)$$

For clarity, we omit the endpoint first-order terms $-\frac{1}{2}(\nabla L(\theta_1)^\top \xi_1 + \nabla L(\theta_2)^\top \xi_2)$, which are negligible when θ_1, θ_2 are sufficiently converged.

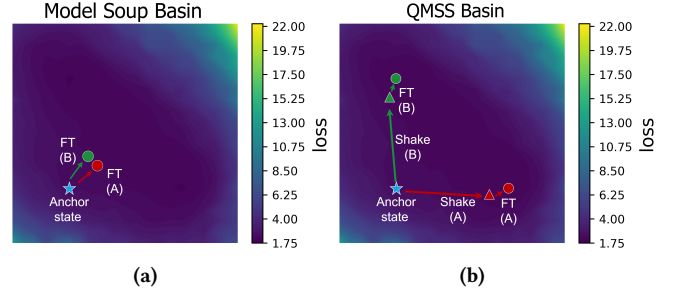


Figure 4: 2D loss landscape visualization of Model Soup vs. QMSS (ResNet-32, CIFAR-100, 4/4-bit). Both plots use the same projection plane for comparison. (a) Standard Model Soup: models fine-tuned from the QAT anchor state (star) collapse to nearly identical positions (circles), showing ensemble degeneracy. (b) QMSS: shake-up perturbations (triangles) yield diverse fine-tuned models (circles) within the same flat basin, preserving LMC while inducing diversity for ensemble gains.

This bound shows that quantization can *amplify* the barrier even when $\mathcal{B}(\theta_1, \theta_2)$ is small, since $\nabla L(\theta_\lambda)$ need not vanish along the segment and the curvature term depends on H_λ .

Why independent QAT can break connectivity. In QAT, quantizer parameters (e.g., step sizes / clipping bounds) are learned jointly with weights, so two independent runs can converge to different effective grids (e.g., $s_1 \neq s_2$), yielding mismatched noise scales in (6). This mismatch amplifies the extra term in (10) and can produce a large interpolation barrier in practice. Fig. 3 confirms this: *No anchor* models show loss spikes near $\lambda \approx 0.5$, indicating disconnected basins.

QAT anchor state: staying in the same basin after convergence. Let θ_0^{anc} be a sufficiently converged QAT anchor state where quantizer parameters have stabilized (e.g., effectively frozen or changing negligibly under small-step fine-tuning). Consider fine-tuning from this state:

$$\theta_i = \theta_0^{\text{anc}} + \Delta_i, \quad i = 1, 2, \quad (11)$$

with small drift $\|\Delta_i\|$ due to SGD noise or minor hyperparameter changes.

If fine-tuning stays within the same local basin around θ_0^{anc} , then by (5), $\mathcal{B}(\theta_1, \theta_2)$ is small. Moreover, because the quantizer grid is shared (or nearly shared) after convergence, $\xi_1, \xi_2, \xi_\lambda$ are aligned (roughly $s_1 \approx s_2 \approx s_\lambda$), reducing grid mismatch. Hence, $\mathcal{B}_Q$ is primarily controlled by the local drift Δ_λ within the basin and the (shared) quantization scale, rather than uncontrolled inter-run quantizer divergence. This explains why a shared *QAT anchor state* preserves LMC among fine-tuned QAT solutions.

Fig. 3 empirically validates this: anchor-based fine-tuning (Model Soup, QMSS) yields near-zero interpolation barriers, whereas independent QAT shows a loss spike near $\lambda \approx 0.5$, indicating disconnected basins. Fig. 4 provides a complementary 2D view, where anchor-based models lie in the same flat basin and high-loss barriers appear only at its periphery.

However, LMC alone is insufficient: standard Model Soup still clusters to nearly identical solutions (Fig. 4a), limiting ensemble gains. We address this in Section 4.

4 Method

Quantized model soup performs weight averaging after fine-tuning a base model θ_0 with quantization-aware training (QAT), using diverse hyperparameter settings following the same recipe as the standard model soup. In this section, we first introduce quantization-aware training and explain our proposed quantized model soup approach with QMSS.

4.1 Quantization-aware training

We follow a hardware-friendly uniform quantization scheme and adopt the EWGS method to optimize the learnable min-max interval. We utilize a uniform quantizer that maps the full-precision value x to a quantized value $\tilde{x}$. The quantization process contains the following steps. First, x is clipped and normalized into the range $[0, 1]$:

$$\tilde{x} = \text{clip}\left(\frac{x-l}{u-l}, 0, 1\right), \quad (12)$$

where $\text{clip}(a, 0, 1) = \min(\max(a, 0), 1)$, and l and u are learnable parameters that differ per layer. For n -bit quantization, the latent value $\tilde{x}$ is converted to a discrete value $\hat{x}$ with a round function:

$$\hat{x} = \frac{\text{round}((2^n - 1)\tilde{x})}{2^n - 1}. \quad (13)$$

Finally, the quantizer generates a quantized weight or activation as:

$$Q(x)_W = \bar{x} = 2(\hat{x} - 0.5), \quad Q(x)_A = \bar{x} = \hat{x}. \quad (14)$$

Here, (14) is expressed in the normalized coordinate induced by (12). In practice, values are rescaled back using (l, u) .

In quantization-aware training (QAT), the round function inside the quantizer inhibits differentiation. To address this, QAT employs the Straight-Through Estimator (STE) by approximating $\frac{\partial \tilde{x}}{\partial x} \approx 1$, and then uses the chain rule to update parameters as

$$x \leftarrow x - \eta \frac{\partial L}{\partial \tilde{x}} \frac{\partial \tilde{x}}{\partial x}. \quad (15)$$

EWGS mitigates the training instability caused by STE by scaling the gradient propagated to the latent value element-wise:

$$\frac{\partial L}{\partial \tilde{x}} = \frac{\partial L}{\partial \hat{x}} \left(1 + \gamma \text{sign}\left(\frac{\partial L}{\partial \tilde{x}}\right) (\tilde{x} - \hat{x})\right), \quad (16)$$

where $\gamma \geq 0$ is a scaling factor.

4.2 Quantized Model Soup

Following the standard model soup recipe, we generate K candidate models by fine-tuning a common base model θ_0 with QAT under different hyperparameter configurations h_i (e.g., learning rate schedules and data augmentations):

$$\theta_i = \text{QAT}(\theta_0, h_i, p_i), \quad i = 1, \dots, K, \quad (17)$$

where p_i optionally specifies a $p\%$ weight perturbation used to promote diversity (described in the next section).

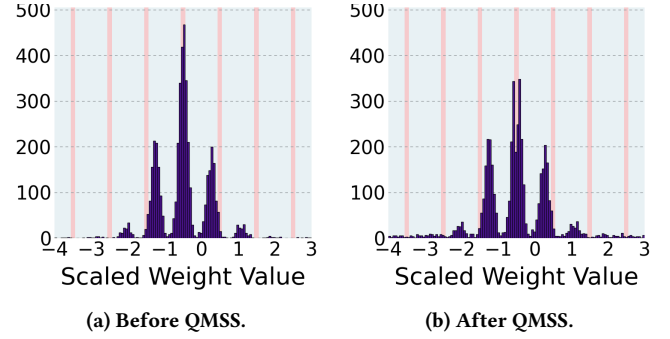


Figure 5: Weight-distribution histograms of a ResNet-32 convolutional layer (a) before QMSS ($p = 0$) and (b) after a 5% QMSS perturbation. Red vertical bands indicate the weights targeted by QMSS, which lie near the quantization thresholds, whereas blue bands denote stable weights unchanged.

Each θ_i can be deployed in its quantized form $\bar{\theta}_i$. To obtain a single merged model, we average candidates in full precision and then quantize the averaged weights:

$$\theta_{\text{soup}} = \frac{1}{K} \sum_{i=1}^K \theta_i, \quad (18)$$

$$\bar{\theta}_{\text{soup}} = Q_w(\theta_{\text{soup}}),$$

where $Q_w(\cdot)$ denotes the weight quantization operator.

A key practical detail is to generate candidates from a **QAT anchor state**, i.e., a sufficiently converged checkpoint where the quantizer parameters have stabilized. When fine-tuning from this anchor, we *freeze* the learnable bounds (l, u) in Eq. 12 so that all candidates share (nearly) the same quantization grid. Otherwise, independently fine-tuned candidates may drift to different bounds, inducing *grid mismatch*, and the final quantization $Q_w(\theta_{\text{soup}})$ can incur a large accuracy drop due to inconsistent bin assignments.

4.3 Quantized Model Soup Shake-Up

In this section, we describe the shake-up method, which introduces a weight perturbation of $p\%$. Let $\theta^\ell \in \mathbb{R}^{d_\ell}$ denote the parameter vector of convolutional layer ℓ , and let $Q(\cdot)$ be the uniform n -bit quantizer defined in Eq. 14. For each weight element $w \in \theta^\ell$, we apply quantization to obtain the quantized weight $\bar{w} = Q(w)_W$.

Unstable-weight detection and layer-wise $p\%$ selection. Let

$$w_q = (2^n - 1) \tilde{x},$$

be the rescaled position of a weight whose normalized value is $\tilde{x}$. The larger $|w_q - \text{round}(w_q)|$ is, the closer w_q lies to the quantization boundary at $\text{round}(w_q) \pm 0.5$. We mark the weight as *unstable* if

$$|w_q - \text{round}(w_q)| \geq \delta. \quad (19)$$

Equivalently, with $\Delta(w_q) = w_q - \text{round}(w_q)$, criterion (19) becomes $|\Delta(w_q)| \geq \delta$. For convolutional layer ℓ we collect all such weights into the set

$$\mathcal{U}_\ell = \left\{ w \in \theta_\ell \mid \left| (2^n - 1) \tilde{x}(w) - \text{round}((2^n - 1) \tilde{x}(w)) \right| \geq \delta \right\}. \quad (20)$$

Only the smallest-magnitude $p\%$ of $\mathcal{U}_\ell$ are subsequently perturbed. We then keep only the smallest-magnitude $p\%$ of these unstable weights, as follows. First, sort their magnitudes in *ascending* order,

$$\{|w_{(1)}| \leq |w_{(2)}| \leq \dots \leq |w_{(|\mathcal{U}_\ell|)}|\}, \quad (21)$$

and define

$$r = \left\lceil \frac{p}{100} |\mathcal{U}_\ell| \right\rceil, \quad T_\ell(p) = |w_{(r)}|.$$

We define the set S_ℓ of weights selected as unstable in the ℓ -th convolutional layer as follows:

$$S_\ell = \{w \in \mathcal{U}_\ell \mid |w| \leq T_\ell(p)\}, \quad (22)$$

i.e., the lowest-magnitude $p\%$ among the weights deemed unstable.

To balance diversity and candidate quality, we restrict perturbations to boundary-near weights and further filter them by low magnitude. This design follows directly from the geometry of uniform quantization. A bounded floating-point perturbation can change the resulting integer code only when a weight lies within a small margin of a quantization threshold, whereas perturbations applied far from thresholds typically map back to the same quantized value and therefore do not induce quantized diversity. The low-magnitude filter preserves candidate strength by avoiding high-saliency weights while still maximizing quantized code flips, as measured by QWS. As shown in Appendix Table 10, this choice yields the strongest candidates and the largest soup gains, along with the highest QWS.

Random perturbation. For every $w \in S_\ell$ we replace its value with one drawn *uniformly* from the convolutional layer’s dynamic range, leaving all remaining weights unchanged:

$$w \leftarrow \text{Unif}(l_\ell, u_\ell), \quad (23)$$

where l_ℓ and u_ℓ are the learnable min–max bounds of layer ℓ (Eq. 12). Fig. 5a shows the QAT model without perturbation ($p = 0$), while Fig. 5b shows the model after a 5% random perturbation. The perturbation shifts the weight distribution and thus changes bin assignments. The overall procedure is summarized in Algorithm 1.

Preserving linear mode connectivity. A natural concern is whether the injected perturbations might push models out of the shared loss basin, breaking the LMC required for effective weight averaging. We argue that QMSS preserves connectivity because perturbations are restricted to a small fraction of low-magnitude weights near quantization boundaries. These weights are inherently unstable due to boundary oscillation and have limited influence on the learned representations (Section 3.2), so modifying them is unlikely to distort the local loss geometry or move solutions across basins.

Fig. 3 supports this claim: QMSS models exhibit near-zero interpolation barriers, comparable to standard Model Soup variants and far below independently trained QAT models. Fig. 4 provides a complementary 2D view. Starting from the anchor state, shake-up perturbations move candidates to distinct intermediate positions. After fine-tuning, they converge to different final locations while remaining within the same flat basin. In contrast, standard Model Soup candidates collapse to nearly identical solutions, illustrating the ensemble degeneracy problem. This dual property, preserving LMC for reliable weight averaging while inducing quantized-domain diversity for ensemble gains, underlies the effectiveness of QMSS.

Algorithm 1 Quantized Model Soup Shake-Up & Soup

Require: Pre-trained QAT model θ_0 with per-layer clip ranges $\{(l_\ell, u_\ell)\}$; perturb ratio p (%); boundary margin δ ; bit-width n ; number of variants K ; fine-tune schedule $\{\eta_t\}_{t=1}^T$.

Ensure: Final quantized soup model $\bar{\theta}_{\text{soup}}$.

```

1:  $\mathcal{M} \leftarrow \emptyset$  {set of perturbed models}
2: for  $i \leftarrow 1$  to  $K$  do
3:    $\theta_i \leftarrow \theta_0$  {clone the seed}
4:   for all conv layer  $\ell$  in  $\theta_i$  do
5:      $U_\ell \leftarrow \emptyset$  {unstable-weight buffer}
6:     for all  $w \in \ell.\text{weight}$  do
7:        $\tilde{x} \leftarrow \text{clip}((w - l_\ell)/(u_\ell - l_\ell), 0, 1)$ 
8:        $w_q \leftarrow (2^n - 1) \tilde{x}$ 
9:       if  $|w_q - \text{round}(w_q)| \geq \delta$  then
10:         $U_\ell \leftarrow U_\ell \cup \{w\}$ 
11:       end if
12:     end for
13:     Sort  $U_\ell$  by  $|w|$  (ascending)
14:      $m \leftarrow \lceil p \cdot |U_\ell| / 100 \rceil$ 
15:      $S_\ell \leftarrow$  first  $m$  elements of  $U_\ell$ 
16:     for all  $w \in S_\ell$  do
17:        $w \sim \mathcal{U}(l_\ell, u_\ell)$  {shake-up}
18:     end for
19:   end for
20:   Freeze all  $\{l_\ell, u_\ell\}$  in  $\theta_i$ 
21:   Fine-tune  $\theta_i$  for  $T$  steps with schedule  $\{\eta_t\}$ 
22:    $\mathcal{M} \leftarrow \mathcal{M} \cup \{\theta_i\}$ 
23: end for
24:  $\theta_{\text{soup}} \leftarrow \frac{1}{K} \sum_{\theta \in \mathcal{M}} \theta$ 
25:  $\bar{\theta}_{\text{soup}} \leftarrow Q_w(\theta_{\text{soup}})$ 
26: return  $\bar{\theta}_{\text{soup}}$ 

```

5 Experiments

5.1 Experimental Setup

Across all tables, the best result is in **bold** and the second best is underlined. We evaluate QMSS on image classification using CIFAR-100 with ResNet-32 [24], Tiny-ImageNet and ImageNet with ResNet-18 [7, 17]. For audio classification, we use FSD-Kaggle2018 [10] with Audio Spectrogram Transformer [14]. Robustness is assessed on CIFAR-100-C and Tiny-ImageNet-C [18]. We primarily use EWGS-based QAT and additionally validate on LSQ [9] to demonstrate generalization. We use fixed default settings for p and δ throughout and do not tune them per dataset or bit-width. Appendix Tables 14 and 15 show that QMSS yields consistent improvements across a wide range of p and δ . We compare against standard Model Soup and Stochastic Weight Averaging (SWA) [20], both starting from the same QAT anchor state. Code and scripts are available in the Appendix. Full hyperparameter settings are provided in Appendix Table ??.

5.2 Main Results

Tables 1 and 2 present our main results. QMSS consistently outperforms standard Model Soup across bit-widths, confirming that it effectively mitigates the Ensemble Degeneracy problem in QAT.

Table 1: Top-1 Accuracy (%) under different W/A (bits) settings for ResNet-18 on Tiny-ImageNet. Each result is based on weight averaging of 30 models fine-tuned with randomly sampled hyperparameter configurations. All experiments are repeated three times and the results are averaged. The bit-widths of weights (W) and activations (A) are represented in the form of W/A.

	Tiny-ImageNet				Tiny-ImageNet-C			
Acc. of	2/2	3/3	4/4	8/8	2/2	3/3	4/4	8/8
Base QAT	59.62	60.49	60.85	60.81	20.67	20.62	21.43	21.34
Best candidate	60.35	61.00	61.14	61.20	20.52	20.81	21.88	21.77
Mean candidate	59.95	60.65	60.90	60.92	20.39	20.44	21.67	21.52
Model Soup	60.62 ± 0.04	61.31 ± 0.03	61.63 ± 0.16	61.45 ± 0.09	21.04 ± 0.14	20.71 ± 0.15	22.75 ± 0.12	22.46 ± 0.15
Best candidate	60.39	60.67	60.94	60.66	20.91	20.42	21.82	21.81
Mean candidate	59.85	60.30	60.75	60.41	20.42	20.29	21.49	21.56
QMSS (1%)	<u>60.72</u> ± 0.03	<u>61.45</u> ± 0.05	<u>61.81</u> ± 0.07	<u>61.68</u> ± 0.03	<u>21.11</u> ± 0.13	<u>21.30</u> ± 0.17	<u>22.94</u> ± 0.20	<u>22.48</u> ± 0.13
Best candidate	60.22	60.79	61.06	60.88	20.85	20.85	21.78	21.82
Mean candidate	59.77	60.15	60.60	60.37	20.39	20.43	21.47	21.62
QMSS (5%)	61.39 ± 0.16	61.89 ± 0.10	62.35 ± 0.13	62.33 ± 0.06	21.77 ± 0.15	21.82 ± 0.17	23.45 ± 0.13	23.55 ± 0.19

Table 2: Top-1 Accuracy (%) under different W/A (bits) settings for ResNet-32 on CIFAR-100. Each result is based on weight averaging of 30 models fine-tuned with randomly sampled hyperparameter configurations. All experiments are repeated three times and the results are averaged.

	CIFAR-100				CIFAR-100-C			
Acc. of	2/2	3/3	4/4	8/8	2/2	3/3	4/4	8/8
Base QAT	63.05	64.37	65.05	65.27	41.99	42.77	43.11	42.99
Best candidate	63.49	64.69	65.24	65.90	<u>42.79</u>	43.09	43.27	43.25
Mean candidate	63.05	64.44	65.07	65.44	42.58	42.90	43.12	43.02
Model Soup	63.13 ± 0.17	64.43 ± 0.08	65.33 ± 0.10	65.59 ± 0.04	42.23 ± 0.14	42.16 ± 0.17	43.15 ± 0.10	43.30 ± 0.12
Best candidate	<u>63.44</u>	64.56	65.11	65.83	42.66	42.98	43.29	43.12
Mean candidate	62.91	63.83	64.74	65.33	42.43	42.46	43.00	42.86
QMSS (1%)	63.17 ± 0.10	64.75 ± 0.13	<u>65.74</u> ± 0.12	<u>66.05</u> ± 0.04	42.35 ± 0.12	<u>43.35</u> ± 0.16	<u>43.43</u> ± 0.11	<u>43.72</u> ± 0.20
Best candidate	63.21	64.40	65.09	65.58	43.15	43.24	43.19	43.32
Mean candidate	61.69	62.60	64.32	64.90	40.42	42.73	41.45	42.71
QMSS (5%)	62.54 ± 0.19	<u>64.70</u> ± 0.21	65.85 ± 0.11	67.13 ± 0.10	42.19 ± 0.12	43.89 ± 0.20	43.83 ± 0.11	45.39 ± 0.15

Accuracy improvements. On Tiny-ImageNet, QMSS improves accuracy by up to +0.88% at 8/8-bit (61.45% $\rightarrow$ 62.33%). On CIFAR-100, QMSS achieves a maximum gain of +1.54% at 8/8-bit (65.59% $\rightarrow$ 67.13%).

Robustness under distribution shift. On corruption benchmarks, QMSS shows even larger gains. CIFAR-100-C improves from 43.30% to 45.39% at 8/8-bit (+2.09%), and Tiny-ImageNet-C improves from 22.46% to 23.55% (+1.09%).

5.3 Comparison with SWA

Table 3 compares QMSS against Stochastic Weight Averaging [20], which averages checkpoints along a single training trajectory without additional training cost. We apply QMSS on top of SWA by performing shake-up at every 10 epochs during the 30-epoch fine-tuning schedule.

Table 3: Comparison with Stochastic Weight Averaging (SWA) on CIFAR-100 (ResNet-32). SWA and SWA+QMSS use the same 30-epoch fine-tuning budget as standard fine-tuning.

	CIFAR-100			
Method	2/2	3/3	4/4	8/8
Base QAT	63.05	64.37	65.05	65.27
Fine-Tuning	63.15	64.44	65.13	65.44
SWA	<u>63.27</u>	<u>64.48</u>	<u>65.21</u>	<u>65.55</u>
SWA + QMSS (1%)	63.38	64.76	65.69	66.27

While SWA provides modest gains over standard fine-tuning (e.g., 65.21% vs 65.13% at 4/4-bit), combining SWA with QMSS yields substantially larger improvements across all bit-widths. At 8/8-bit, SWA+QMSS achieves 66.27%, outperforming fine-tuning by +0.83%

Table 4: Mean pairwise cosine similarity among candidate models at different bit-widths for CIFAR-100 and Tiny-ImageNet.

Dataset	Method	2/2	4/4	8/8
CIFAR-100	Model Soup	0.996	0.997	0.992
	QMSS (1%)	0.977	0.971	0.925
	QMSS (5%)	0.905	0.882	0.659
Tiny-ImageNet	Model Soup	0.964	0.957	0.954
	QMSS (1%)	0.956	0.946	0.944
	QMSS (5%)	0.876	0.856	0.851

Table 5: Quantized Weight Shift (QWS) for CIFAR-100 and Tiny-ImageNet across bit-width settings.

Dataset	Method	2/2	4/4	8/8
CIFAR-100	Model Soup	1.24%	8.83%	15.93%
	QMSS (1%)	2.24%	10.69%	23.23%
	QMSS (5%)	3.80%	25.69%	41.78%
Tiny-ImageNet	Model Soup	15.84%	36.25%	85.89%
	QMSS (1%)	20.90%	38.75%	86.88%
	QMSS (5%)	27.09%	54.66%	92.81%

and SWA alone by +0.72%. This demonstrates that QMSS provides complementary benefits to temporal weight averaging by inducing structural diversity in the quantized domain.

5.4 Diversity Analysis

We analyze why QMSS improves ensemble performance through diversity metrics.

Limitation of cosine similarity. Cosine similarity is a common proxy for model diversity in weight space [4, 43]. Table 4 shows that QMSS reduces pairwise cosine similarity compared to Model Soup. However, cosine similarity operates in full-precision space and may not capture diversity that persists after quantization, as small floating-point differences can map to the same integer bin.

Quantized Weight Shift (QWS). To directly measure diversity in the quantized domain, we introduce QWS, defined as the mean normalized Hamming distance [15] between each candidate model and the weight-averaged soup:

$$QWS = \frac{1}{K} \sum_{i=1}^K \frac{\|Q(\theta_i) - Q(\theta_{\text{soup}})\|_0}{d} \quad (24)$$

where d is the total number of weights. Table 5 shows that QMSS achieves substantially higher QWS than Model Soup across all settings.

In the Appendix, we provide additional analyses on QWS–accuracy correlation, sensitivity to the boundary threshold and perturbation ratio, and the effect of ensemble size.

Correlation with performance. QWS correlates strongly with ensemble gains (Appendix Table 13). Notably, at 2/2-bit on CIFAR-100,

Table 6: Top-1 Accuracy (%) of QMSS under LSQ quantization across W/A (bits) settings. Results are reported for (left) ResNet-32 on CIFAR-100 using weight averaging of 30 models, and (right) Audio Spectrogram Transformer (AST, ViT-based) on FSD-Kaggle2018 using weight averaging of 10 models. All experiments are repeated three times and the results are averaged.

LSQ + ResNet			LSQ+AST (ViT)		
Method	4/4	8/8	Method	4/8	8/8
Base QAT	66.29	66.12	Base QAT	90.91	91.79
Model Soup	66.27	66.23	Model Soup	91.25	92.01
QMSS (1%)	66.46	66.52	QMSS (1%)	91.69	92.46

Table 7: Top-1 Accuracy (%) under different bit-width settings for ResNet-18 on ImageNet. Each result is based on weight averaging of 20 models fine-tuned with randomly sampled hyperparameter configurations.

ImageNet				
Method	Acc. of	3/3	4/4	8/8
Model Soup	Base QAT	69.34	70.04	70.34
	Best candidate	69.38	70.20	70.40
	Mean candidate	69.21	70.09	70.38
QMSS (1%)	Soup	69.38	70.29	70.42
	Best candidate	69.32	70.15	70.33
	Mean candidate	69.11	69.83	70.23
QMSS (5%)	Soup	69.54	70.43	70.50
	Best candidate	69.11	69.79	70.23
	Mean candidate	68.71	69.57	70.18
	Soup	69.44	70.35	70.65

QWS remains low even with QMSS (3.80% vs. 1.24% for Model Soup), consistent with the limited gains at this extreme bit-width.

Fig. 2 visualizes this relationship. Panel (a) shows that QMSS substantially increases QWS compared to standard Model Soup, confirming greater diversity in the quantized weight space. Panel (b) confirms that QMSS performance improves steadily with ensemble size while Model Soup saturates early.

Additional diversity results are provided in Appendix Table 11.

5.5 Generalization

Table 6 (left) shows that QMSS consistently improves over Model Soup when switching the QAT formulation from EWGS to LSQ, indicating that our gains are not tied to a specific quantizer or surrogate-gradient choice. On CIFAR-100, QMSS improves by +0.19% at 4/4-bit and +0.29% at 8/8-bit. Table 6 (right) further extends the evaluation to a non-vision modality. On Audio Spectrogram Transformer with FSD-Kaggle2018, QMSS achieves larger gains of +0.44% at 4/8-bit and +0.45% at 8/8-bit, demonstrating that the proposed shake-up mechanism transfers beyond CNNs and image classification. Finally, Table 7 reports ImageNet results. QMSS yields consistent improvements across bit-widths, reaching +0.23% at 8/8-bit, which

Table 8: Inference on the Jetson Orin Nano with dataset-specific backbones.

Dataset (Backbone)	Bit	Inference Time (ms)	Speedup	Accuracy (%)
Tiny-ImageNet (ResNet-18)	FP32	2.9621	-	60.50
	INT8	0.4652	6.37×	62.33
CIFAR-100 (ResNet-32)	FP32	0.6742	-	65.58
	INT8	0.3920	1.72×	67.13
ImageNet (ResNet-18)	FP32	3.9881	-	69.76
	INT8	0.5265	7.57×	70.65

is notable given the strong, high-accuracy baselines in this regime. While the absolute gains are smaller than on CIFAR-100, they remain systematic, suggesting that ImageNet models already exhibit higher intrinsic diversity due to larger capacity and scale.

5.6 Inference Speedup

We measure inference latency on a Jetson Orin Nano using TensorRT [32]. As summarized in Table 8, QMSS with INT8 quantization achieves up to 7.57× speedup over FP32 while simultaneously improving accuracy on all evaluated datasets.

5.7 Hyperparameters and Training Cost

QMSS introduces two method-specific hyperparameters: the perturbation ratio p and the boundary margin δ . The complete training hyperparameters are provided in Appendix Table ??; here we focus on the sensitivity of these two QMSS-specific choices. In all main experiments, we use fixed default values and do not tune them per dataset or bit-width. As shown in Appendix Tables 14 and 15, QMSS remains effective across a wide range of p and δ , indicating that the method is not overly sensitive to these choices. In practice, we find $p = 5\%$ to work well when a sufficiently large candidate pool is available, while $p = 1\%$ provides a conservative choice when fewer candidates are used or when candidate quality is more critical.

QMSS also requires additional offline training to generate multiple soup candidates. This overhead should be interpreted as a structural trade-off of the model soup and model merging paradigm, rather than as an additional inference-time burden introduced by QMSS. As in standard model soup, multiple candidates are produced once during offline training, while the final deployed model is obtained by weight averaging and remains a single quantized network. Thus, QMSS is not designed to be training-free; instead, it trades one-time candidate-generation cost for single-model deployment. Let N_{QAT} denote the number of epochs used to train the QAT anchor, N_{rec} denote the recovery fine-tuning epochs for each QMSS candidate, and K denote the number of candidates. When the QAT anchor training cost is normalized as 1×, the sequential training cost of QMSS is

$$1 + \frac{KN_{\text{rec}}}{N_{\text{QAT}}}.$$

For CIFAR-100, this corresponds to $N_{\text{QAT}} = 400$, $N_{\text{rec}} = 30$, and $K = 30$, yielding 3.25× total sequential training cost. In concrete

Table 9: Cost–accuracy trade-off on CIFAR-100 with ResNet-32. Training and inference costs are normalized to a single Base QAT model.

Bit-width	Method	Training Cost	Inference Cost	Accuracy (%)
4/4	Base QAT	1×	1×	65.05
	Deep Ensemble ($K = 3$)	3×	3×	66.12
	Deep Ensemble ($K = 5$)	5×	5×	66.64
	LPE-BSR ($S = 10$)	1×	10×	65.32
	LPE-BSR ($S = 20$)	1×	20×	65.74
	QMSS (ours)	3.25×	1×	65.85
8/8	Base QAT	1×	1×	65.27
	Deep Ensemble ($K = 3$)	3×	3×	67.72
	Deep Ensemble ($K = 5$)	5×	5×	68.41
	LPE-BSR ($S = 10$)	1×	10×	66.15
	LPE-BSR ($S = 20$)	1×	20×	66.98
	QMSS (ours)	3.25×	1×	67.13

terms, on a single RTX 3090, anchor training takes approximately 2.4 hours and the 30 fine-tuning runs take approximately 5.5 hours sequentially, for a total of 7.9 hours. Since candidate generation is independent, these runs can be parallelized when resources are available, although the total compute remains unchanged.

For a smaller offline budget, SWA+QMSS offers a lower-cost operating point by applying shake-up within the same 30-epoch fine-tuning budget as SWA (Table 3). This suggests that QMSS can be used either as full candidate generation or as lightweight diversity injection under a fixed fine-tuning budget.

Table 9 further contextualizes this training cost against output-level ensemble methods. Deep Ensemble [25] can achieve higher accuracy by evaluating multiple independently trained models, but its inference cost increases linearly with the number of members. Similarly, LPE-BSR [30] improves performance through multiple inference-time samples. In contrast, QMSS averages candidates in weight space and deploys only one quantized model. Therefore, the practical trade-off is different: output-level ensembles spend additional cost at every inference, whereas QMSS spends additional cost once offline and preserves 1× inference cost, memory footprint, and deployed parameter count. This distinction is particularly important for quantization and on-device deployment, where latency, memory, and energy are often more restrictive than offline training cost.

6 Conclusion

We proposed QMSS, a method that overcomes ensemble degeneracy in QAT by injecting targeted perturbations to boundary-near, low-magnitude weights. Our analysis shows that fine-tuning from a converged QAT anchor state preserves linear mode connectivity, yet the discrete quantizer collapses model diversity. QMSS resolves this by inducing quantized-domain diversity while remaining within the same loss basin. Across multiple benchmarks, two quantizers, and two architecture families, QMSS consistently outperforms standard model soup with no additional inference cost.

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Table 10: Ablation on perturbation location and candidate filter (CIFAR-100, ResNet-32, EWGS, 4/4-bit). “Boundary-near” selects weights within a δ -margin of quantization thresholds. “Low-magnitude” restricts perturbations to a small-magnitude subset to preserve candidate strength. QWS denotes the mean normalized Hamming distance between each candidate and the weight-averaged soup (higher is more diverse in the quantized domain).

Location	Candidate filter	Note	Best cand.	Mean cand.	Soup	Cos. sim.	QWS
Random	Random		63.80	61.65	64.02	0.860	21.97%
Random	Low-magnitude		64.12	63.08	64.11	0.819	24.40%
Boundary-near	Random		63.74	62.89	64.71	0.811	20.74%
Boundary-near	Low-magnitude	QMSS	65.09	64.32	65.85	0.882	25.69%

A Ablation: Perturbation Location and Candidate Filter

QMSS is designed to induce *quantized-domain* diversity without sacrificing candidate strength. Rather than injecting generic random noise, QMSS applies *quantization-aware* perturbations by selecting weights within a δ -margin of quantization thresholds, where small full-precision (FP) changes are most likely to flip the assigned integer bin. In addition, perturbations are restricted to a low-magnitude subset to preserve candidate quality while still encouraging bin-level diversity.

This design is not an ad-hoc heuristic, but follows directly from the structure of uniform quantization. To create *effective* diversity after quantization, perturbations must translate into different integer assignments, which is most likely for boundary-near weights; perturbing far-from-boundary weights often remains invisible after quantization (no bin change). At the same time, indiscriminately flipping many weights can harm candidate quality and potentially distort the local geometry. Filtering by low magnitude serves as a safety mechanism: it biases perturbations toward parameters with smaller functional influence, enabling bin-level diversity while keeping candidates within the same basin and preserving mergeability.

Table 10 reports an ablation on CIFAR-100 4/4-bit (ResNet-32, EWGS), varying (i) where candidate weights are chosen (random vs. boundary-near) and (ii) how candidates are filtered (random vs. low-magnitude). All strategies induce non-trivial diversity signals (QWS increases; cosine similarity varies), yet their outcomes differ. Random perturbations tend to increase diversity but often degrade candidate quality, limiting soup gains. Boundary-near selection improves merging but may still weaken candidates when applied broadly. Combining boundary-near selection with the low-magnitude filter (QMSS) yields the best balance, producing the strongest candidates and the largest soup improvement together with the highest QWS. Overall, these results support that the gains arise from *quantization-aware perturbations* that increase *mergeable quantized-domain diversity*, rather than arbitrary noise injection.

B Additional Experiments

B.1 EQ-diversity [41].

Table 11 shows that QMSS achieves lower (better) FQ-BD, FQ-KW, and FQ-GD scores than Model Soup, indicating increased ensemble diversity. These metrics capture complementary aspects of diversity among candidates and are designed to reflect how well a candidate

Table 11: Comparison of EQ-diversity metric scores ($\downarrow$ is better) on 4/4-bit quantized ResNet weight averaging of 30 models.

Dataset	Method	FQ-BD	FQ-KW	FQ-GD
CIFAR-100	Model Soup	0.8943	0.9489	0.9407
	QMSS (1%)	0.8703	0.9373	0.9251
	QMSS (5%)	0.8357	0.9206	0.9010
Tiny-ImageNet	Model Soup	0.8999	0.9516	0.9444
	QMSS (1%)	0.8937	0.9485	0.9423
	QMSS (5%)	0.8696	0.9370	0.9252

set covers diverse yet performant solutions. The consistent reductions suggest that QMSS does not merely inject random noise, but produces a candidate pool with meaningfully different solutions that remain amenable to aggregation.

Importantly, this observation aligns with our quantized-domain analysis: by increasing bin-level diversity (higher QWS), QMSS also improves functional diversity measured by EQ-diversity. In contrast, standard Model Soup can exhibit high similarity after quantization, which limits the effective diversity available for ensembling even when candidates differ slightly in full precision.

Table 12: Comparison of EQ-diversity metric scores ($\downarrow$ indicates better ensemble selection performance) between Model Soup and QMSS. All results on CIFAR-100 are based on 4/4-bit quantized ResNet-32 models.

Dataset	m	Method	FQ-BD	FQ-KW	FQ-GD
CIFAR-100	3	Model Soup	0.8979	0.9660	0.9451
		QMSS (1%)	0.8668	0.9556	0.9267
		QMSS (5%)	0.8368	0.9303	0.9021
	10	Model Soup	0.8949	0.9527	0.9417
		QMSS (1%)	0.8717	0.9423	0.9268
		QMSS (5%)	0.8340	0.9253	0.9014
	30	Model Soup	0.8943	0.9489	0.9407
		QMSS (1%)	0.8703	0.9373	0.9251
		QMSS (5%)	0.8357	0.9206	0.9010

Table 13: Spearman’s correlation r and p-value (entries with $p < 0.05$ are bolded). Across both datasets, QWS shows a stronger positive correlation with ensemble accuracy than $1 - \cos$.

Dataset	Metric	Spearman r	p-value
CIFAR-100	$1 - \cos$	0.5333	0.1392
	QWS	0.7500	0.0199
Tiny-ImageNet	$1 - \cos$	0.5833	0.0992
	QWS	0.7333	0.0246

B.2 EQ-Diversity vs. Number of QMSS Models

We further study how EQ-diversity changes as the number of averaged models increases. Table 12 reports EQ-diversity scores for Model Soup and QMSS with $m \in \{3, 10, 30\}$ models on CIFAR-100 (4/4-bit, ResNet-32). QMSS consistently yields lower FQ-BD, FQ-KW, and FQ-GD across all m , suggesting that the diversity gains persist as the ensemble size grows.

As m increases, Model Soup exhibits only marginal changes in EQ-diversity, consistent with candidate collapse in the quantized space. By contrast, QMSS maintains a stronger diversity signal even for larger m , indicating that additional candidates contribute non-redundant information rather than averaging out to the same solution. This helps explain why QMSS continues to improve with ensemble size, whereas Model Soup tends to saturate early despite adding more candidates. Overall, these results support that QMSS produces a scalable candidate set whose diversity is both measurable (via EQ-diversity) and beneficial for weight-space merging.

B.3 Correlation Between QWS and Accuracy

To assess whether similarity measures are truly correlated with ensemble accuracy, we use Spearman’s rank correlation instead of Pearson’s correlation. This choice is motivated by the characteristics of our data: cosine similarity lies in $[-1, 1]$, whereas QWS lies in $[0, 1]$, so their marginal distributions are not directly comparable. Moreover, given the small sample size ($n = 9$) and potential non-linear or non-normal patterns, a rank-based measure is more robust. Spearman’s correlation evaluates monotonic relationships based on ranks rather than absolute values, and is less sensitive to outliers. Table 13 reports Spearman’s correlation values for cosine similarity and QWS on CIFAR-100 and Tiny-ImageNet.

B.4 Effect of Boundary Threshold and Perturbation Ratio

Unless otherwise noted, we use a fixed boundary threshold across all experiments: $\delta = 0.40$ for QMSS (1%) and $\delta = 0.35$ for QMSS (5%). To assess sensitivity, we vary the perturbation ratio p and the boundary threshold δ on CIFAR-100 (ResNet-32, 4/4-bit). As shown in Table 14 and Table 15, QMSS remains consistently effective across a wide range of p and δ , indicating that the method is not overly sensitive to these hyperparameters. In particular, performance stays competitive even away from the best setting, suggesting that QMSS generalizes well without per-dataset or per-bit tuning.

Table 14: Perturbation ratio p (%) vs. Top-1 accuracy on CIFAR-100 (ResNet-32, 4/4-bit).

	10%	7%	5%	3%	1%
Best candidate	64.75	64.89	65.09	65.24	65.11
Mean candidate	62.64	63.84	64.32	64.61	64.74
QMSS	65.66	65.74	65.85	65.30	65.74

Table 15: Boundary δ vs. Top-1 accuracy on CIFAR-100 (ResNet-32, 4/4-bit).

	0	0.1	0.2	0.3	0.35	0.4
QMSS (1%)	65.39	65.37	65.44	65.38	65.47	65.74
QMSS (5%)	65.33	65.47	65.65	65.74	65.85	65.68

B.5 Environment

Hardware. All AST (ViT-based) experiments were run on an NVIDIA A100 GPU, while all other experiments (CIFAR-100, Tiny-ImageNet, ImageNet) were run on an NVIDIA RTX 3090.

Datasets & splits. For CIFAR-100, Tiny-ImageNet, and ImageNet, we reserve 10% of the *training* set as a validation set for model selection; the official test (or validation, in the case of ImageNet) set is never used for tuning and is only used for final reporting.

Code & reproducibility. An anonymized code repository is available at: anonymous.4open.science link

C Diagnostic Study: Do Single-Model Quantization Techniques Resolve Ensemble Degeneracy?

Several concurrent quantization methods aim to improve the accuracy or robustness of a *single* quantized model. In contrast, our setting targets *weight-space merging* (Model Soup) under QAT, where the main failure mode is *ensemble degeneracy*: independently fine-tuned candidates can collapse to nearly identical quantized representations, yielding limited merge gains. To clarify this distinction and address missing-related-work concerns, we perform a diagnostic study asking a focused question: *when adapted as candidate-generation recipes, do these mechanisms induce mergeable quantized diversity?*

Protocol. On CIFAR-100 with ResNet-32 at 4/4-bit, we instantiate intent-matched, simplified versions of representative mechanisms and then apply the same candidate-generation and weight-space merging pipeline used throughout the paper (i.e., identical anchor-based fine-tuning and soup construction). We report (i) the best and mean candidate accuracy, (ii) the final soup accuracy, and (iii) diversity indicators in both full precision (cosine similarity) and the quantized domain (QWS; higher is better).

Adapted mechanisms. **Tang-style [37] (weight coupling):** add a one-shot weight-coupling regularization step before generating multiple QAT candidates and merging.

An-style [2] (dynamic/static pruning): apply a pruning-based

Table 16: Diagnostic comparison with intent-matched adaptations of related mechanisms (CIFAR-100, ResNet-32, 4/4-bit). All methods use the same anchor-based candidate-generation and weight-space merging pipeline; the listed techniques are adapted as candidate-generation *recipes* rather than treated as direct baselines. Lower cosine similarity indicates higher full-precision diversity, while higher QWS indicates higher quantized-domain diversity.

Method	Best cand.	Mean cand.	Soup	Cos. sim. ↓	QWS ↑
Tang	65.45	65.17	65.32	0.999	1.16%
An	65.48	65.11	65.27	0.998	5.62%
Zhang	65.12	64.74	64.25	0.996	6.01%
Wang	65.42	65.06	65.07	0.993	5.08%
QMSS	65.09	64.32	65.85	0.882	25.69%

initialization (zeroing a fixed fraction of low-magnitude weights) before QAT, then generate candidates and merge.

Zhang-style [45] (MagR): apply magnitude reduction prior to QAT to mitigate outlier-driven range expansion, then generate candidates and merge.

Wang-style [38] (NPFT): apply sensitivity-guided perturbation fine-tuning to improve robustness of a single quantized model, then generate candidates and merge.

Findings. As shown in Table 16, these adapted baselines preserve strong single-model accuracy, but yield only small quantized diversity ($\text{QWS} \approx 1\text{--}6\%$) and correspondingly limited soup gains. In contrast, QMSS substantially increases QWS (25.69%) and achieves the largest improvement after merging. This supports our central claim that addressing ensemble degeneracy requires explicitly inducing *quantized-domain* diversity while preserving linear mode connectivity.